

OLS $\rightarrow$ Ordinary Least Solⁿ
 $\hookrightarrow$ closed form Solⁿ

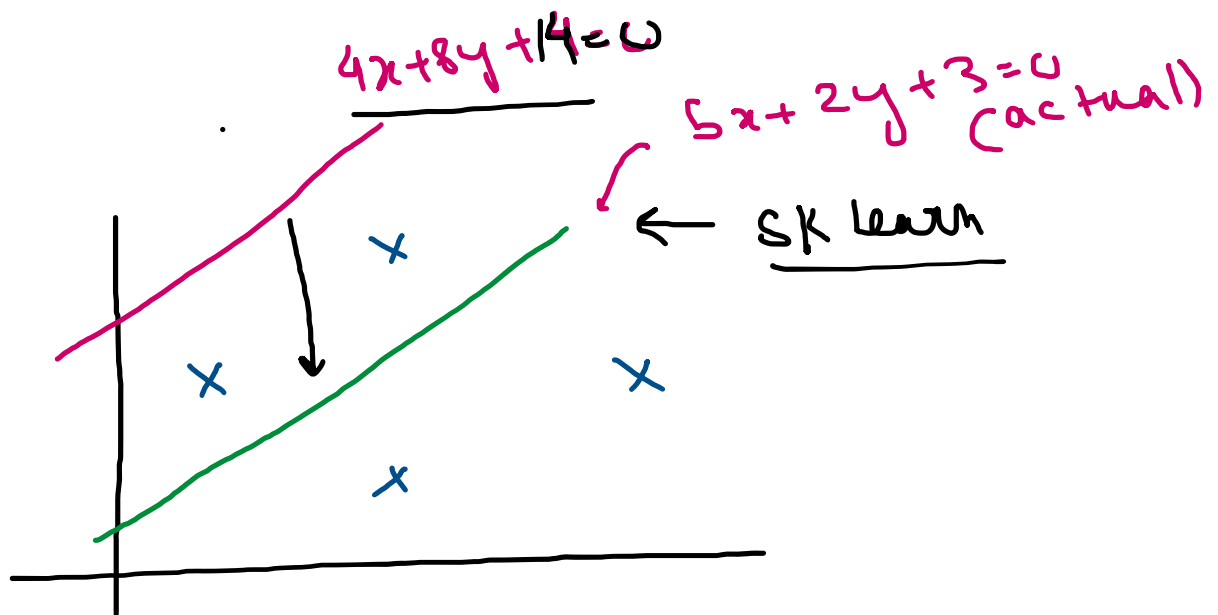
$$\left[\frac{dL}{dw} = 0 \right]$$

Disadvantages:-

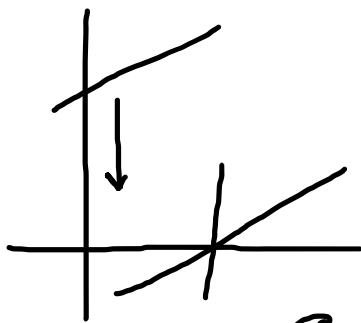
$$\frac{dL}{dw} = Y X^T \cdot \underbrace{\underbrace{(X^T X)^{-1}}_{\sim \Theta(n^2)}}_{MLR}$$

row x col

1. computational expense
2. slow



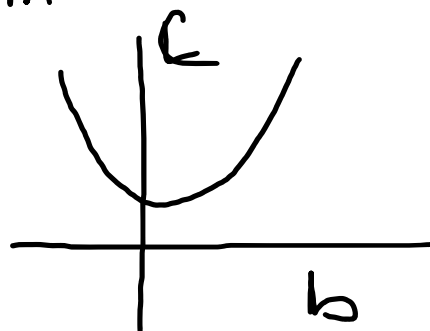
$4x + 8y + 14 = 0 \longrightarrow 5x + 2y + 3 = 0$
 + transformations
 (m, b)



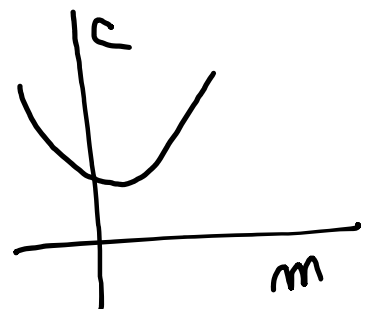
$$C = \sum_{i=1}^m (y_i - mx_i - b)^2 \quad \leftarrow m, b$$

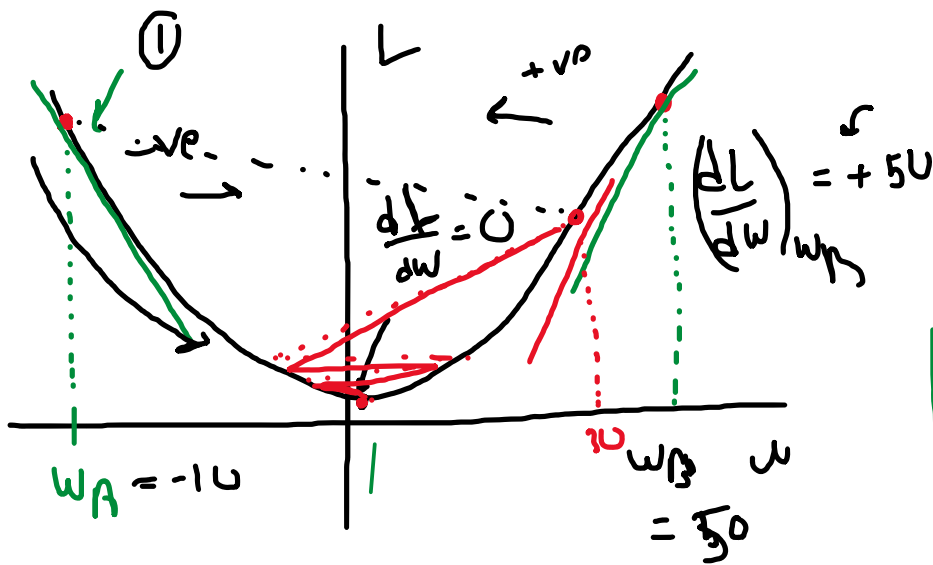
$$\frac{1}{n} C$$

$m \rightarrow \text{const.}$



$b \rightarrow \text{const.}$





$$\left(\frac{dL}{dw}\right)_{w=w_A} = -40$$

$$w_A - \left[\frac{dL}{dw}\right]_{w=w_A}$$

$$-10 - (-40) = 30$$

$$w_{new} = w_{old} - \left[\frac{dL}{dw}\right]_{w=w_{old}}$$

$$y = mx + b$$

$$m_{new} = m_{old} - \frac{dL}{dm_{old}}$$

$$b_{new} = b_{old} - \frac{dL}{db_{old}}$$

$$w_0 + w_1 x_1 + w_2 x_2 + \dots + w_n x_n = 0$$

$$\frac{dL}{dw} = 0$$

$$y = x^2$$

$$\left(\frac{dy}{dx}\right)_{x=2}$$

$$\frac{dy}{dx} = 2x$$

$$: 2 \cdot 2 = 4$$

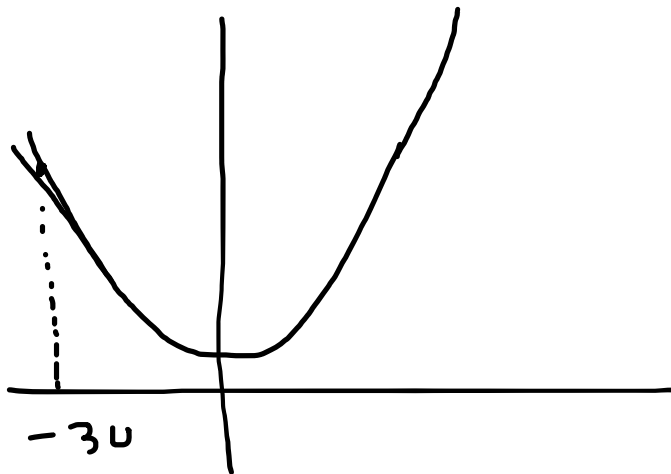
w_0
 w_1

w_r

learning rate

0.01

$$w_n = w_0 - \eta \frac{dL}{dw_0}$$



$$w = -30$$

$$\frac{dL}{dw_{30}} = -10$$

$$w = -30 - (0.01)(-10)$$
$$= -30 + 0.1$$

$$w = -29.9$$

η ← learning rate
impact

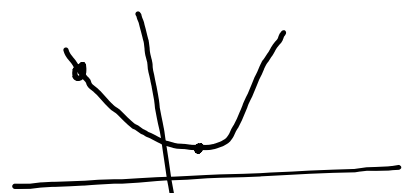
$$0 \leq \eta \leq 1$$

$$w_n = w_0 - \eta \frac{dL}{dw_0}$$

overshoot

$$\eta = 1$$

$$w_n = w_0 - \frac{dL}{dw_0}$$



convergence

①

$$w_{\text{new}} - w_{\text{old}} \rightarrow 0$$

$$\text{① epoch} \leftarrow 100 \quad \checkmark$$

for i in range(epoch)
... $\eta \frac{dL}{dw}$

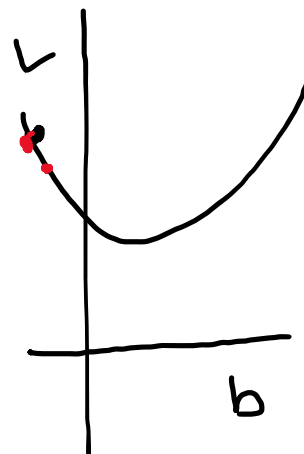
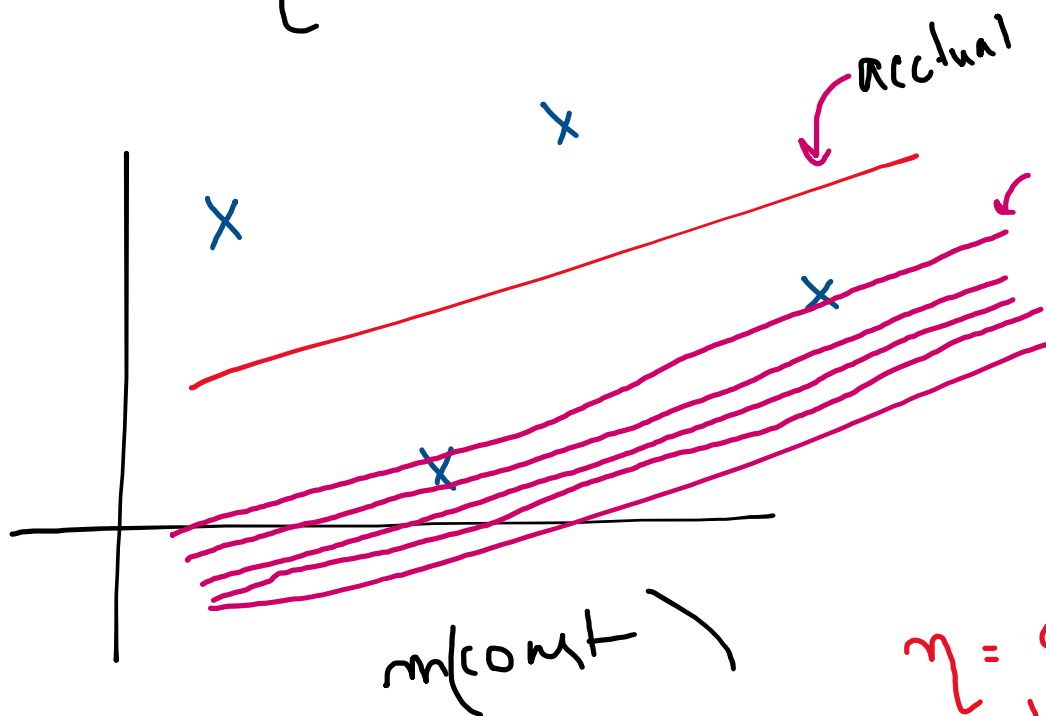
$$\eta = 1$$

$$W_n = W_0 \leftarrow$$

$$W_n = W_0 + \eta \frac{dL}{dW_0}$$

$$0 < \eta < 1$$

$$\eta \leftrightarrow \text{epoch}$$



$$\text{epoch} = \frac{4}{14}$$

$$\eta = 0.0001$$

$$W_n = W_0 - \eta \frac{dL}{dW_0}$$

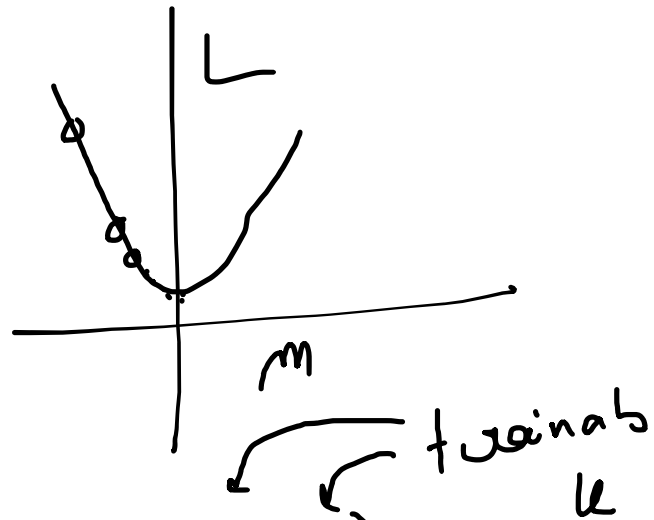
$$y : mx + b$$

$$W_n = W_0 - \eta \frac{dL}{dW_0}$$

$$LR \rightarrow L = \frac{1}{n} \sum_{i=1}^n (y_i - mx_i - b)$$

$$\frac{dL}{dw_{old}} = \frac{dL}{dm_{old}}$$

$$\frac{dL}{db_{old}}$$



$$b_n = b_0 - \eta \frac{dL}{db_0}$$

$$m_n = m_0 - \eta \frac{dL}{dm_0}$$

$$L = f(w, b)$$

$$SLR = f(m, b)$$

$$MLR = f(w_0, w_1, w_2, \dots)$$

Simple LR

$$L = \sum (y_i - mx_i - b)^2 \quad \text{MSK}$$

$$m_n = m_0 - \eta \frac{dL}{dm_0}$$

$$b_n = b_0 - \eta \frac{dL}{db_0}$$

$$\frac{dL}{dm_0} = \frac{d}{dm_0} \sum (y_i - mx_i - b)^2, \quad \left| \frac{dL}{db_0} = \frac{d}{db_0} \sum (y_i - mx_i - b)^2 \right.$$

$$\frac{d}{dm_0} = \frac{d}{dm_1} \left(\frac{1}{2} \right)$$

$$= \frac{1}{2}$$

$$= -2(y_i - m x_i - b) x_i$$

$$= -2(y_i - m x_i - b)$$

Simple Linear Regression:-

$$y = mx + b \Rightarrow w_1 x + w_0$$

MLR.

$$y = w_0 + w_1 x_1 + w_2 x_2 + \dots + w_n x_n \quad \begin{matrix} n \text{ cols} \\ \rightarrow (n+1) \end{matrix}$$

$$= \sum_{i=0}^n x_i w_i$$

$W = W - \eta \frac{dL}{dw}$

Cost function.

$$w_1 = w_1 - \eta \frac{dL}{dw_1}$$

$$w_2 = w_2 - \eta \frac{dL}{dw_2}$$

$$w_n = w_n - \eta \frac{dL}{dw_n}$$

$$w_0 = w_0 - \eta \frac{dL}{dw_0}$$

$\xrightarrow{n}$
 $x_1 \quad x_2 \quad \dots \quad x_n$

MSE

$$L = \frac{1}{n} \sum_{i=1}^m (y_i - \hat{y}_i)^2$$

$$L = \frac{1}{n} (y_1 - \hat{y}_1)^2 + (y_2 - \hat{y}_2)^2 + \dots + (y_m - \hat{y}_m)^2$$

$$L = \frac{1}{n} (y_1 - w_0 - w_1 x_{11} - w_2 x_{12} - \dots - w_n x_{1n})^2 + \dots$$

$$+ \frac{1}{n} (y_2 - w_0 - w_1 x_{21} - w_2 x_{22} - \dots - w_n x_{2n})^2 + \dots$$

$$+ \frac{1}{n} (y_m - w_0 - w_1 x_{m1} - w_2 x_{m2} - \dots - w_n x_{mn})^2$$

=

$$+ \frac{1}{n} (y_{m-1} w_0 - w_1 x_{m-1} - w_2 x_{m-2} - \dots - w_n x_{m-n})$$

$$\frac{dL}{dw_0} \quad \frac{dL}{dw_1} \quad \frac{dL}{dw_2} \quad \dots \quad \frac{dL}{dw_n}$$

$$\frac{dL}{dw_0} = \frac{1}{n} \left[(y_1 - \hat{y}_1) \cdot (-1) + 2(y_2 - \hat{y}_2) \cdot (-1) + \dots + 2(y_{m-1} - \hat{y}_{m-1}) \cdot (-1) \right]$$

$$= -\frac{2}{n} \left[(y_1 - \hat{y}_1) + (y_2 - \hat{y}_2) + \dots + (y_{m-1} - \hat{y}_{m-1}) \right]$$

$$= -\frac{2}{n} \sum_{i=1}^m (y_i - \hat{y}_i) \left| \begin{array}{ccc} y & \hat{y} & y - \hat{y} \\ \hline & & \\ \hline & & \\ \hline & & \\ \hline & & \\ \hline & & \end{array} \right\}$$

$$\frac{dL}{dw_0} = -\frac{2}{n} \sum_{i=1}^m (y_i - \hat{y}_i) \leftarrow x_{i0}$$

$$\frac{dL}{dw_1} = \frac{2}{n} (y_1 - \hat{y}_1)^2 \cdot (-x_{11}) + 2(y_2 - \hat{y}_2) \cdot (-x_{21}) + \dots + \frac{2}{n} (y_{m-1} - \hat{y}_{m-1}) \cdot (-x_{m-1,1})$$

$$= -\frac{2}{n} \sum_{i=1}^m (y_i - \hat{y}_i) x_{i1} \quad \left| \begin{array}{c} x_1 \\ x_2 \\ \vdots \\ x_n \end{array} \right| x_2 - x_n y \hat{y}$$

$$= -2 \sum_{i=1}^m (y_i - \hat{y}_i) x_{i1}$$

$$\frac{dL}{dw_1} = -2 \sum_{i=1}^m (y_i - \hat{y}_i) x_{i1}$$

$$\frac{dL}{dw_2} = -2 \sum_{i=1}^m (y_i - \hat{y}_i) x_{i2}$$

$$\frac{dL}{dw_n} = -2 \sum_{i=1}^m (y_i - \hat{y}_i) x_{in}$$

$$= -2 \sum_{i=1}^m (y_i - \hat{y}_i) x_{in}$$

x_1
 x_2
 $\vdots$
 x_n

$$\hat{y}_1 = w_0 + w_1 x_{11} + w_2 x_{12} + \dots + w_n x_{1n}$$

$$\hat{y}_2 = w_0 + w_1 x_{21} + w_2 x_{22} + \dots + w_n x_{2n}$$

$$\vdots$$

$$\hat{y}_m = w_0 + w_1 x_{m1} + w_2 x_{m2} + \dots + w_n x_{mn}$$

$$\hat{Y} = \begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \vdots \end{bmatrix} = \begin{bmatrix} w_0 + w_1 x_{11} + w_2 x_{12} + \dots + w_n x_{1n} \\ w_0 + w_1 x_{21} + w_2 x_{22} + \dots + w_n x_{2n} \\ \vdots \end{bmatrix}$$

$$Y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{bmatrix} = \begin{bmatrix} w_0 + w_1 x_{11} + w_2 x_{12} \dots \\ \vdots \\ w_0 + w_1 x_{m1} + w_2 x_{m2} \dots + w_n x_{mn} \end{bmatrix}$$

$$\begin{bmatrix} 1 & x_{11} & x_{12} & \dots & x_{1n} \\ 1 & x_{21} & x_{22} & \dots & x_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{m1} & x_{m2} & \dots & x_{mn} \end{bmatrix} \begin{bmatrix} w_0 \\ w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix} \quad \leftarrow Y \quad (I)$$

$$w_0 + \begin{bmatrix} x_{11} & x_{12} & \dots & x_{1n} \\ x_{21} & x_{22} & \dots & x_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ x_{m1} & x_{m2} & \dots & x_{mn} \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix} \quad \leftarrow Y \quad (II)$$

$$W = \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix}$$

$$W_{\text{-der}} = \begin{bmatrix} \frac{dL}{dw_1} \\ \frac{dL}{dw_2} \\ \vdots \\ \frac{dL}{dw_n} \end{bmatrix}$$

$$W_{\text{der}} = \begin{bmatrix} \frac{dL}{dw_1} \\ \frac{dL}{dw_2} \\ \vdots \\ \frac{dL}{dw_n} \end{bmatrix} = \begin{bmatrix} -\frac{2}{m}(Y - \hat{Y}) \cdot x_1 \\ -\frac{2}{m}(Y - \hat{Y}) \cdot x_2 \\ \vdots \\ -\frac{2}{m}(Y - \hat{Y}) \cdot x_n \end{bmatrix}$$

$$= -\frac{2}{m}$$

$$X = \begin{bmatrix} [x_{11}, x_{12}, x_{13}, \dots, x_{1m}] \\ [x_{21}, x_{22}, x_{23}, \dots, x_{2n}] \\ \vdots \\ [x_{m1}, x_{m2}, x_{m3}, \dots, x_{mn}] \end{bmatrix}$$

$$W = \begin{bmatrix} w_1 & w_2 & w_3 & \dots & w_n \end{bmatrix}$$

$$\begin{aligned}
 \frac{dL}{dW} &= \left[\frac{dL}{dw_1} \quad \frac{dL}{dw_2} \quad \dots \quad \frac{dL}{dw_n} \right] \\
 &= -\frac{2}{n} \left[(Y - \hat{Y})x_1 \quad (Y - \hat{Y})x_2 \quad \dots \quad (Y - \hat{Y})x_n \right] \\
 &= -\frac{2}{n} (Y - \hat{Y}) \left[x_1, x_2, \dots, x_n \right] \\
 &= \boxed{-\frac{2}{n} (Y - \hat{Y}) X^T} \quad \left| \begin{array}{l} [x_{11}, x_{12}, x_{13}, \dots, x_{1n}], \\ [x_{21}, x_{22}, x_{23}, \dots, x_{2n}], \\ \vdots \\ [x_{m1}, x_{m2}, x_{m3}, \dots, x_{mn}] \end{array} \right. \\
 &\quad \left[\begin{array}{ccc} x_{11} & x_{12} & x_{1n} \\ x_{21} & x_{22} & x_{2n} \\ \vdots & \vdots & \vdots \\ x_{m1} & x_{m2} & x_{mn} \end{array} \right] \\
 &\quad \left. \right]
 \end{aligned}$$

Problem

24 October 2025 21:37

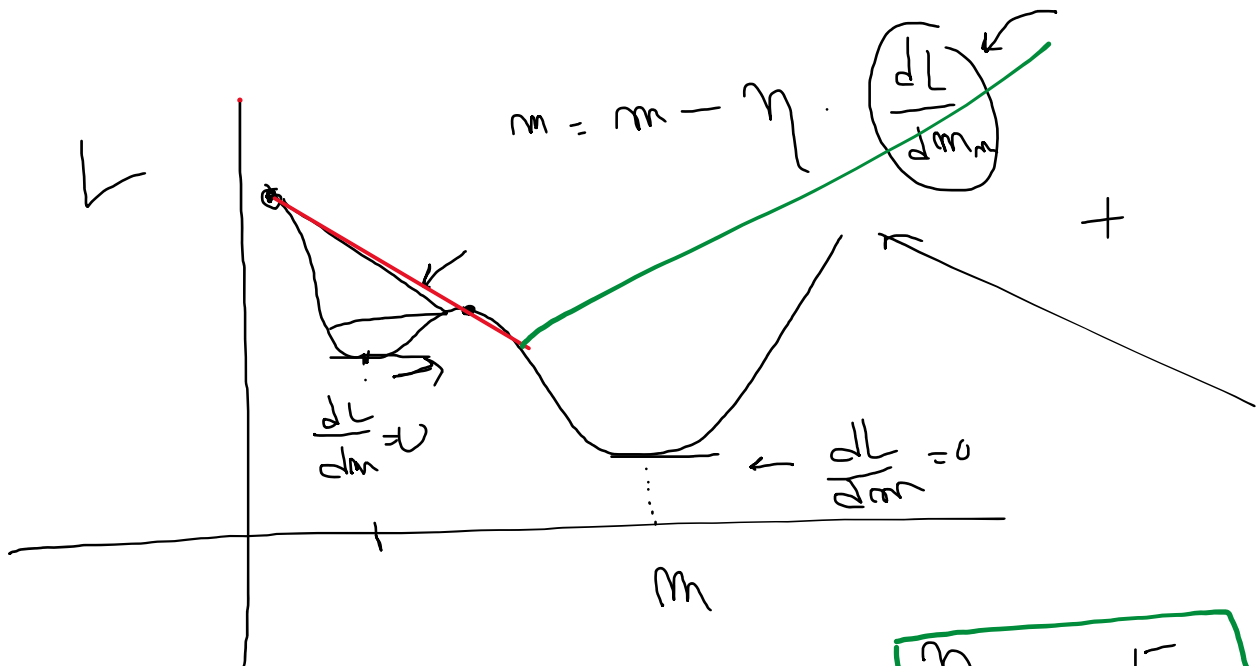
```
class OurGradientDescent:
    def __init__(self, learning_rate = 0.01 , epochs = 100):
        self.lr = learning_rate
        self.epochs = epochs
        self.weights = None
        self.intercept = None

    def fit(self, X, y):
        n_samples, n_features = X.shape
        # Initialization
        self.intercept = 0
        self.weights = np.zeros(n_features)
        print(self.weights.shape)

        # Gradient Descent
        for i in range(self.epochs):
            y_cap = self.intercept + np.dot(X, self.weights)
            i_der = (-2 / n_samples) * np.sum(y - y_cap)
            self.intercept = self.intercept - self.lr * i_der
            w_der = (-2 / n_samples) * np.dot(X.T , (y - y_cap) )
            self.weights = self.weights - self.lr * w_der

    def predict(self,X):
        return np.dot(X, self.weights) + self.intercept
```

Data Size
>
RAM
?



$$m = -40$$

$$\left(\frac{dL}{dm} \right)_{-40} = -10$$

$$\eta = 5$$

$$0 < \eta < 1$$

$$(-40 - 5(-10))$$

$$m = -40 - 5(-10) = 10$$

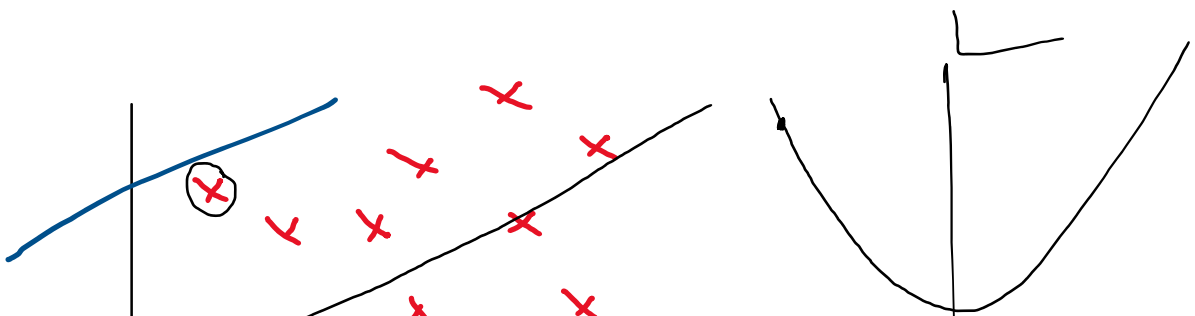
$$0 < \eta < 1$$

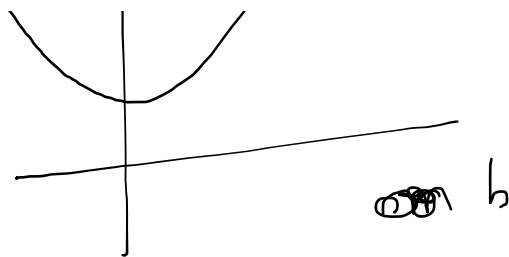
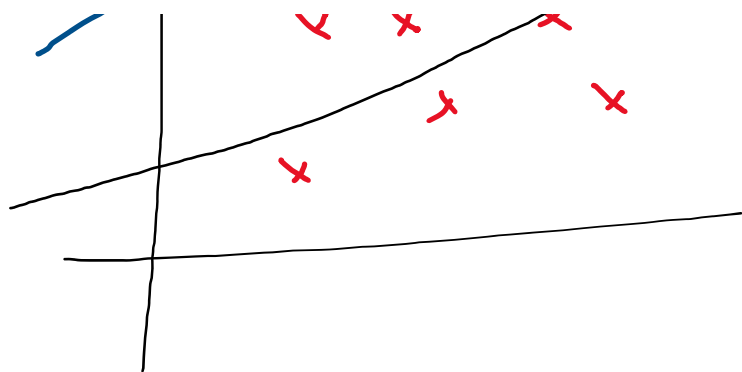
$$\frac{\partial L}{\partial w_n} = -\frac{2}{m} (Y - \hat{Y}) X_n$$

$$\frac{\partial L}{\partial W} = \begin{bmatrix} \frac{\partial L}{\partial w_1} \\ \vdots \\ \frac{\partial L}{\partial w_n} \end{bmatrix} = -\frac{2}{m} (Y - \hat{Y}) X$$

RAM !!

memory issues





~~not a cost~~

$m \neq \text{const}$

$$m = m - \eta \frac{dL}{dm}$$

$$= m - \eta \frac{d}{dm} (y_j - mx_j - b)^2$$

$$= m - \eta 2 \cdot (y_j - \hat{y}_j) (-x_j)$$

$$m = m + 2\eta (y_j - \hat{y}_j) x_j$$

$$b = b + 2\eta (y_j - \hat{y}_j)$$

$$b = b - \eta \frac{dL}{db}$$

$$\hat{y} = mx + b$$

$$L = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

$$= \frac{1}{n} \sum_{i=1}^n (y_i - mx_i - b)^2$$

$$L = \sum (y_i - mx_i - b)^2$$

random $\rightarrow j$

$$L = (y_j - \hat{y}_j)^2$$

$$L = (y_j - mx_j - b)^2$$

$x_1 \quad x_2 \quad x_3 \quad \dots \quad x_m \quad y \quad y$
 1
 2
 3
 :
 m

$w_0 \quad w_1 \dots w_m$

$$w_0 = w_0 - \eta \frac{dL}{dw_0} \quad \left| \quad L = (y_j - \hat{y}_j)^2 \right.$$

$$w_0 = w_0 - \eta \frac{d}{dw_0} \left(y_j - \hat{y}_j \right)^2 \cdot (-1) = (y_j - w_0 - w_1 x_{j1} - w_2 x_{j2} - \dots - w_n x_{jn})^2$$

$$\boxed{w_0 = w_0 + 2\eta (y_j - \hat{y}_j)}$$

$$w_1 = w_1 - \eta \frac{dL}{dw_1}$$

$$= w_1 - \eta \frac{d}{dw_1} (y_j - \hat{y}_j)^2 \cdot (-x_{j1})$$

$$\boxed{w_1 = w_1 + 2\eta (y_j - \hat{y}_j) x_{j1}}$$

$$\boxed{w_2 = w_2 + 2\eta (y_j - \hat{y}_j) x_{j2}}$$

$$w_2 = \dots$$

$$w_n = w_n + 2\eta (y_i - \hat{y}_i) x_{jn}$$

for i in epochs:-
 for k in X.shape[0]:
 j = random

$$\hat{y}_j = w_0 + w_1 x_{j1} + w_2 x_{j2} + \dots + w_n x_{jn}$$

$$= w_0 + (X_j \cdot W)$$

$$\frac{dL}{dW} = \left[\frac{dL}{dw_1} \quad \frac{dL}{dw_2} \quad \dots \quad \frac{dL}{dw_n} \right]$$

=

$$\frac{dL}{dw_1} = \frac{d}{dw_1} (y_j - \hat{y}_j)^2$$

$$\frac{dL}{dw_1} = \frac{d}{dw_1} (y_j - \hat{y}_j)$$

$$= \frac{d}{dw_1} (y_j - w_0 - w_1 x_{j1} - w_2 x_{j2} - \dots - w_n x_{jn})^2$$

$$= 2 (y_j - \hat{y}_j) (-x_{j1})$$

$$\frac{dL}{dw_1} = -2 (y_j - \hat{y}_j) (x_{j1})$$

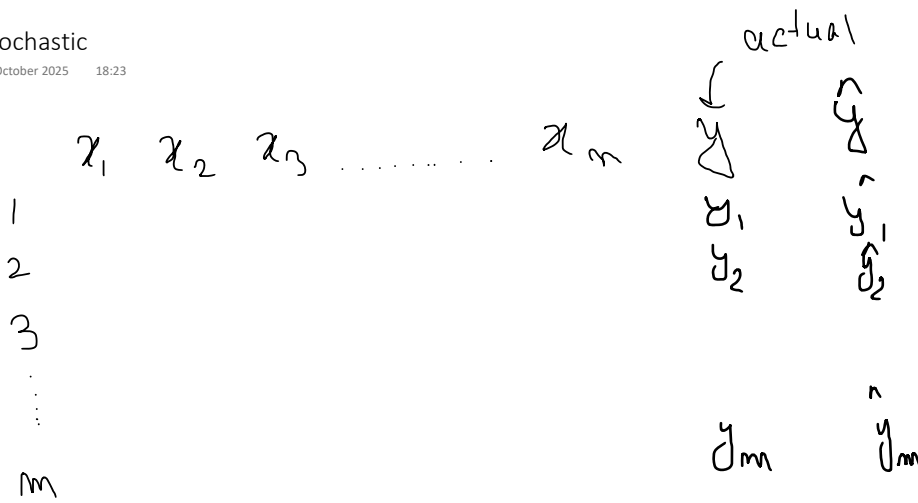
$$\frac{dL}{dw_2} = -2 (y_j - \hat{y}_j) (x_{j2})$$

$$\frac{dL}{dw_n} = -2 (y_j - \hat{y}_j) (x_{jn})$$

$$\frac{dL}{dw} = \begin{bmatrix} \frac{dL}{dw_1} & \frac{dL}{dw_2} & \dots & \frac{dL}{dw_n} \end{bmatrix}$$

$$= -2 (y_j - \hat{y}_j) \begin{bmatrix} x_{j1} & x_{j2} & \dots & x_{jn} \end{bmatrix}$$

$$\boxed{\frac{dL}{dw} = -2 (y_j - \hat{y}_j) X_j}$$



MLR

$$\hat{y}_1 = w_0 + w_1 x_{11} + w_2 x_{12} + \dots + w_m x_{1m} \quad [1 \times 1]$$

$$= w_0 + [x_{11} \ x_{12} \ \dots \ x_{1m}] \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_m \end{bmatrix} \quad [n \times m]$$

$1 \times m$ $m \times 1$

$$\hat{y}_1 = w_0 + x_1 \cdot w$$

$$= \hat{y}_1 = \sum_{i=0}^m w_i x_{i1} \quad \checkmark \quad x_{01} = 1$$

$$\Rightarrow \hat{y}_2 = w_0 + x_2 w$$

$$\Rightarrow \hat{y}_m = w_0 + x_m w$$

$$\hat{y} = \begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \vdots \\ \hat{y}_m \end{bmatrix}$$

$$\begin{aligned}
 & \begin{matrix} n \\ \downarrow \\ (m, 1) \\ (1, m) \\ \hline \end{matrix} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{bmatrix} \cdot \begin{bmatrix} w_0 + w_1 x_{11} + w_2 x_{12} + \dots + w_n x_{1m} \\ w_0 + w_1 x_{21} + w_2 x_{22} + \dots + w_n x_{2m} \\ \vdots \\ w_0 + w_1 x_{m1} + w_2 x_{m2} + \dots + w_n x_{mn} \end{bmatrix} \\
 & \begin{bmatrix} (m, 1) \\ (1, m) \\ \hline \end{bmatrix} = \underline{w_0} + \begin{bmatrix} x_{11} & x_{12} & \dots & x_{1m} \\ x_{21} & x_{22} & \dots & x_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ x_{m1} & x_{m2} & \dots & x_{mn} \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix}
 \end{aligned}$$

$$= \boxed{w_0 + X \cdot \underline{w^T}}$$

$$\downarrow \quad \underline{m \times 1}$$

$$= (w_0 + X \cdot w^T)^T$$

$1 \times m$

①

$$\underline{(m \times n)} \times \underline{(n \times 1)}$$

②

$$\begin{array}{ccc}
 \begin{matrix} m \times n \\ \downarrow \\ m \times 1 \end{matrix} & \xleftrightarrow{\text{Comp}} & \begin{matrix} m \times 1 \\ \downarrow \\ 1 \times n \end{matrix} \\
 & & \underline{1 \times n}
 \end{array}$$

③

$$w_0 + w \cdot x$$

$$\downarrow$$

$$(1 \times n) \times (n \times m)$$

$$\downarrow +$$

$$(1 \times m) \quad (m \times m)$$

$$= (1 \times m)$$

$$\text{Coeff} - der = \frac{dL}{dw_0}$$

$$L = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

$$= \frac{1}{n} \sum_{i=1}^n (y_i - w_0 - w_1 x_{i1} - w_2 x_{i2} \dots - w_m x_{in})^2$$

$$= \frac{1}{n} \sum_{i=1}^n 2(y_i - \hat{y}_i)(-1)$$

$$= -\frac{2}{n} \sum_{i=1}^n (y_i - \hat{y}_i)$$

$$= -2 \cdot \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)$$

$$= -2 \cdot \frac{1}{n} \left[(y_1 - \hat{y}_1) + (y_2 - \hat{y}_2) + \dots + (y_n - \hat{y}_n) \right]$$

$$= -2 \cdot \frac{Y - \hat{Y}}{n}$$

$$w_1, w_2, \dots = \left[\frac{dL}{dw_1}, \frac{dL}{dw_2}, \dots, \frac{dL}{dw_n} \right]$$

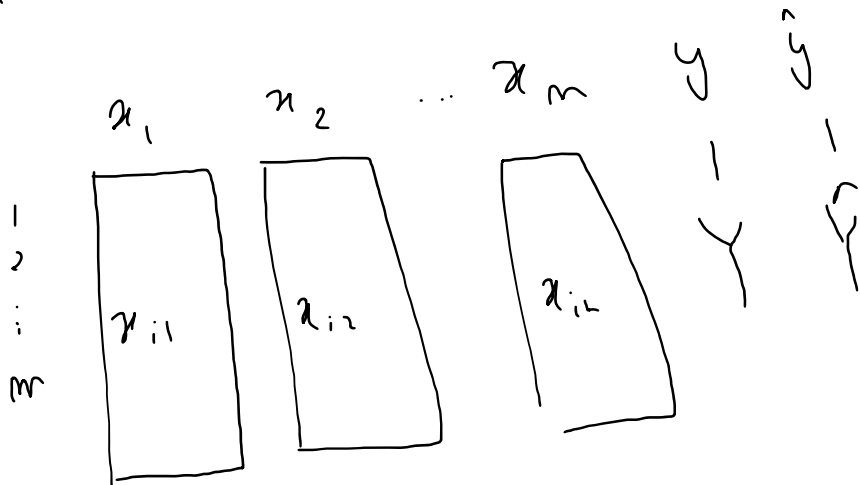
$$w_{\text{des}} = \left[\frac{dL}{dw_1} \quad \frac{dL}{dw_2} \quad \dots \quad \frac{dL}{dw_n} \right]$$

$$\begin{aligned} \frac{dL}{dw_1} &= \frac{d}{dw_1} \left(\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \right) \\ &= \frac{d}{dw_1} \left(\frac{1}{n} \sum_{i=1}^n (y_i - w_0 - w_1 x_{i1} - w_2 x_{i2} - \dots - w_n x_{in})^2 \right) \\ &= \frac{1}{n} \sum_{i=1}^n 2(y_i - \hat{y}_i)(-x_{i1}) \end{aligned}$$

$$\frac{dL}{dw_1} = -\frac{2}{n} \sum_{i=1}^n (y_i - \hat{y}_i) x_{i1} =$$

$$\frac{dL}{dw_2} = -\frac{2}{n} \sum_{i=1}^n (y_i - \hat{y}_i) x_{i2}$$

$$\frac{dL}{dw_n} = -\frac{2}{n} \sum_{i=1}^n (y_i - \hat{y}_i) x_{in}$$



$$\begin{aligned}
 \frac{dL}{dw_1} &= -\frac{2}{n} \sum_{i=1}^n (y_i - \hat{y}_i) x_{i1} \\
 &= -\frac{2}{n} \left[(y_1 - \hat{y}_1) x_{11} + (y_2 - \hat{y}_2) x_{21} + \dots + (y_m - \hat{y}_m) x_{m1} \right] \\
 &= -\frac{2}{n} \begin{bmatrix} (y_1 - \hat{y}_1) & (y_2 - \hat{y}_2) & \dots & (y_m - \hat{y}_m) \end{bmatrix} \begin{bmatrix} x_{11} \\ x_{21} \\ \vdots \\ x_{m1} \end{bmatrix}
 \end{aligned}$$

$$\frac{dL}{dw_1} = \boxed{-\frac{2}{n} (Y - \hat{Y}) \cdot x_1}$$

$$\frac{dL}{dw_2} = -\frac{2}{n} (Y - \hat{Y}) \cdot x_2$$

$$\vdots \\
 \frac{dL}{dw_n} = -\frac{2}{n} (Y - \hat{Y}) \cdot x_n$$

$$w_{\text{den}} = \left[\frac{dL}{dw_1} \quad \frac{dL}{dw_2} \quad \dots \quad \frac{dL}{dw_n} \right]$$

$$= \left[-\frac{2}{n} (Y - \hat{Y}) x_1 \quad -\frac{2}{n} (Y - \hat{Y}) x_2 \quad \dots \quad -\frac{2}{n} (Y - \hat{Y}) x_n \right]$$

$$= \left[-\frac{2}{n} \frac{(1 \dots 1)}{n} \right]$$

$$= -\frac{2}{n} (Y - \hat{Y}) \begin{bmatrix} x_1 & x_2 & \dots & x_n \end{bmatrix}$$

$$\begin{bmatrix} \lambda_1 & \lambda_2 \\ \vdots & \vdots \end{bmatrix}$$

$$= -2 \frac{(Y - \hat{Y})}{n} \begin{bmatrix} [x_{11}, x_{12}, x_{13}, \dots, x_{1m}], \\ [x_{21}, x_{22}, x_{23}, \dots, x_{2m}], \\ \vdots \\ [x_{1m}, x_{2m}, \dots, x_{mm}] \end{bmatrix}$$

$$w_{den} = \boxed{-2 \frac{(Y - \hat{Y})}{n} \cdot x^*}$$

-2

(442) (442,10)

11

(1,442) (442 x 10)

[1, 10]

(353,1)

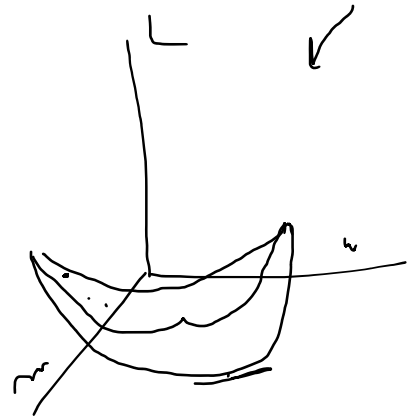
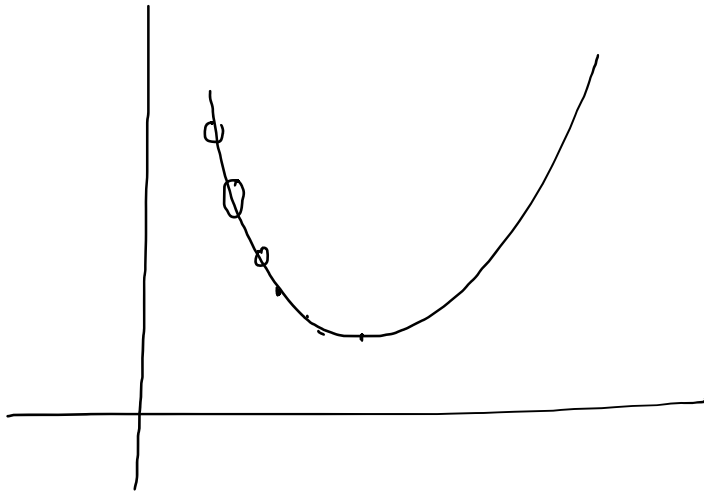
(353,10)

/

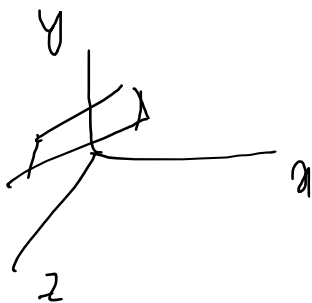
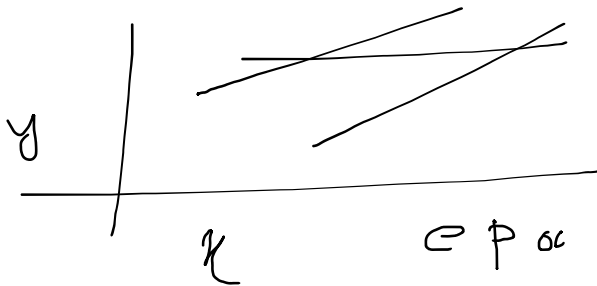
1353,10

$$(1, 353) \times (353, 1)$$

$$= 1116$$



$$L = f(w, w)$$



1. Randomness

Same w of epoch.

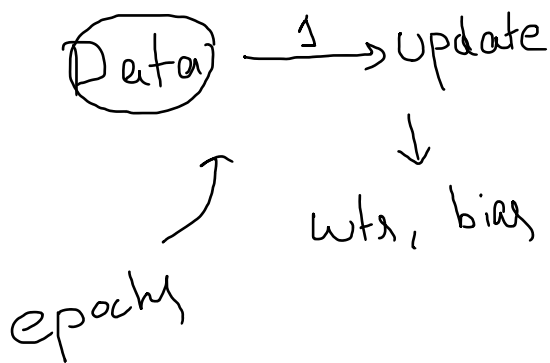
Same nr of epoch.
↙ faster? ↘

BGD
✓
① loop

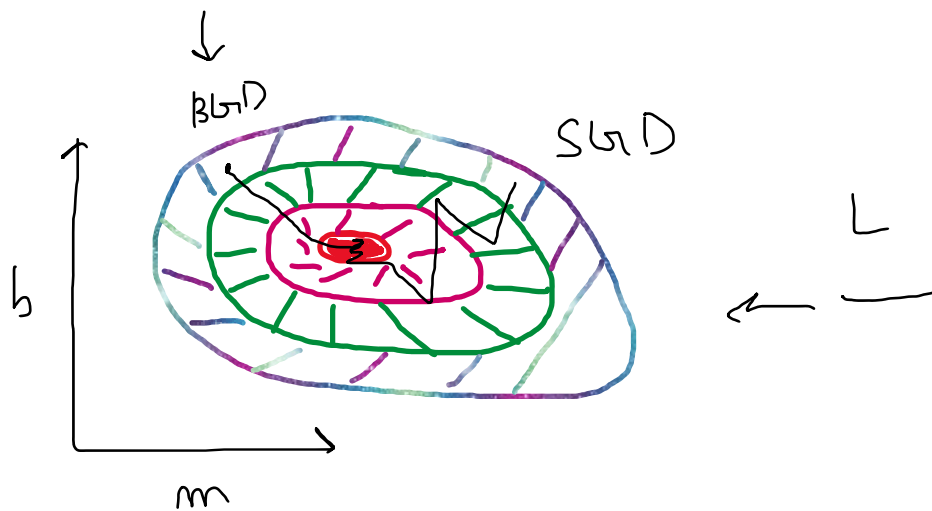
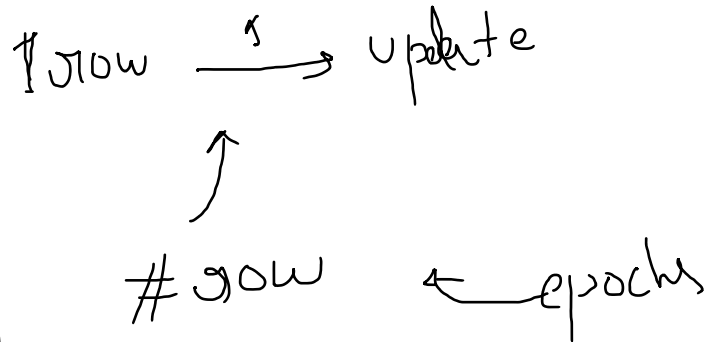
SGD
② loop

Overall faster → SGD

Batch GD



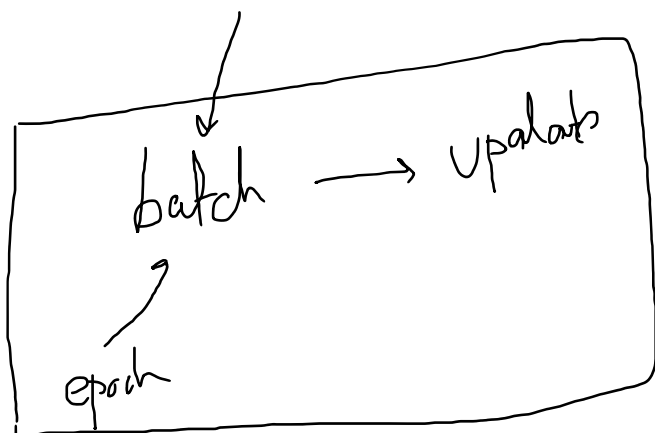
SGD



Mini batch GD: — to reduce the stochasticity.

$x_1, x_2, \dots, x_n, y, \hat{y}$
 1
 2
 3
 4
 ⋮
 m

data $\rightarrow$ batch
 $m \sim \text{row}$
 $1 \text{ batch} = 10 \text{ row}$
 $\left(\frac{m}{10}\right) \leftarrow \text{batch}$



Batch GD (Stoc) Cost

$$w_0 = w_0 - \eta \frac{dL}{dw_0}$$

$$L = \frac{1}{n} \sum_{i=0}^m (y_i - \hat{y}_i)^2$$

$$w_0 = w_0 - \eta \frac{d}{dw_0} \frac{1}{b} \sum_{i=1}^b (y_i - \hat{y}_i)^2$$

$$\text{SGD} = L = (y_i - \hat{y}_i)^2$$

$$= w_0 - \eta \sum_{i=1}^b (y_i - \hat{y}_i) (-1)$$

$$\text{MGN} \quad L = \frac{1}{b} \sum_{i=1}^b (y_i - \hat{y}_i)^2$$

$$= w_0 + \frac{2\eta}{b} \sum (y_i - \hat{y}_i)$$

$$= \frac{1}{b} \sum_{i=1}^b (y_i - w_0 - w_1 x_{i1} - w_2 x_{i2} - \dots - w_n x_{in})^2$$

$$(w_0 = \underline{w_0} + \frac{2\eta}{b} \sum (y - \hat{y}))$$

$$\hat{y} = \left[w_0 + w_1 x_1 + w_2 x_2 \right]$$

$$\hat{y} = w_0 + w_1 x_1 + w_2 x_2 + \dots + w_n x_n$$

$$\hat{y} = w_0 + x[j] w$$

$$\frac{dL}{dw_1} = \frac{d}{dw_1} \frac{1}{b} \sum_{i=1}^b (y_i - w_0 - w_1 x_{i1} - w_2 x_{i2} - \dots - w_n x_{in})^2$$

$$\frac{dL}{dw_1} = \frac{1}{b} 2 \sum_{i=1}^b (y_i - \hat{y}_i) x_{i1} = -\frac{2}{b} (Y - \bar{Y}) X_1$$

$$\frac{dL}{dw} = \left[\frac{dL}{dw_1} \quad \frac{dL}{dw_2} \quad \dots \quad \frac{dL}{dw_n} \right]$$

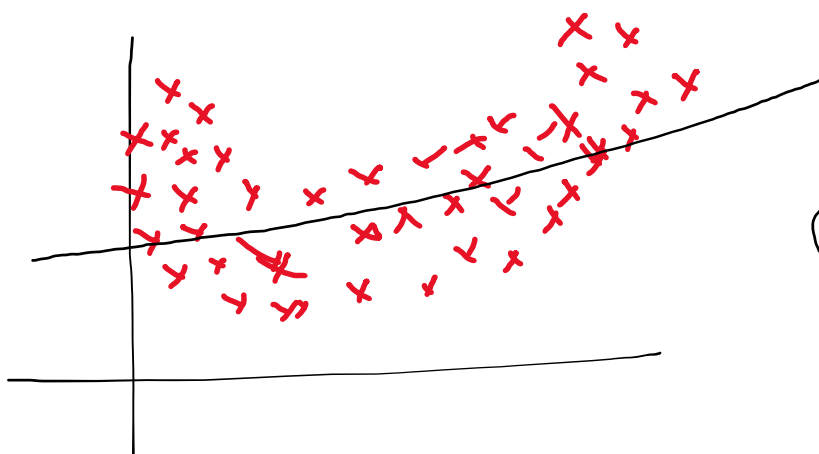
$$= \left[-\frac{2}{b} (Y - \bar{Y}) x_1 \quad -\frac{2}{b} (Y - \bar{Y}) x_2 \quad \dots \quad -\frac{2}{b} (Y - \bar{Y}) x_n \right]$$

$$= -\frac{2}{b} (Y - \bar{Y}) \begin{bmatrix} x_1 & x_2 & \dots & x_n \end{bmatrix}$$

$$= -\frac{2}{b} \underline{(Y - \bar{Y})} \underline{X[j]}$$

Polynomial Regression

27 October 2025 20:46



$$y = w_0 + w_1x_1 + w_2x_2 + \dots + w_nx_n$$